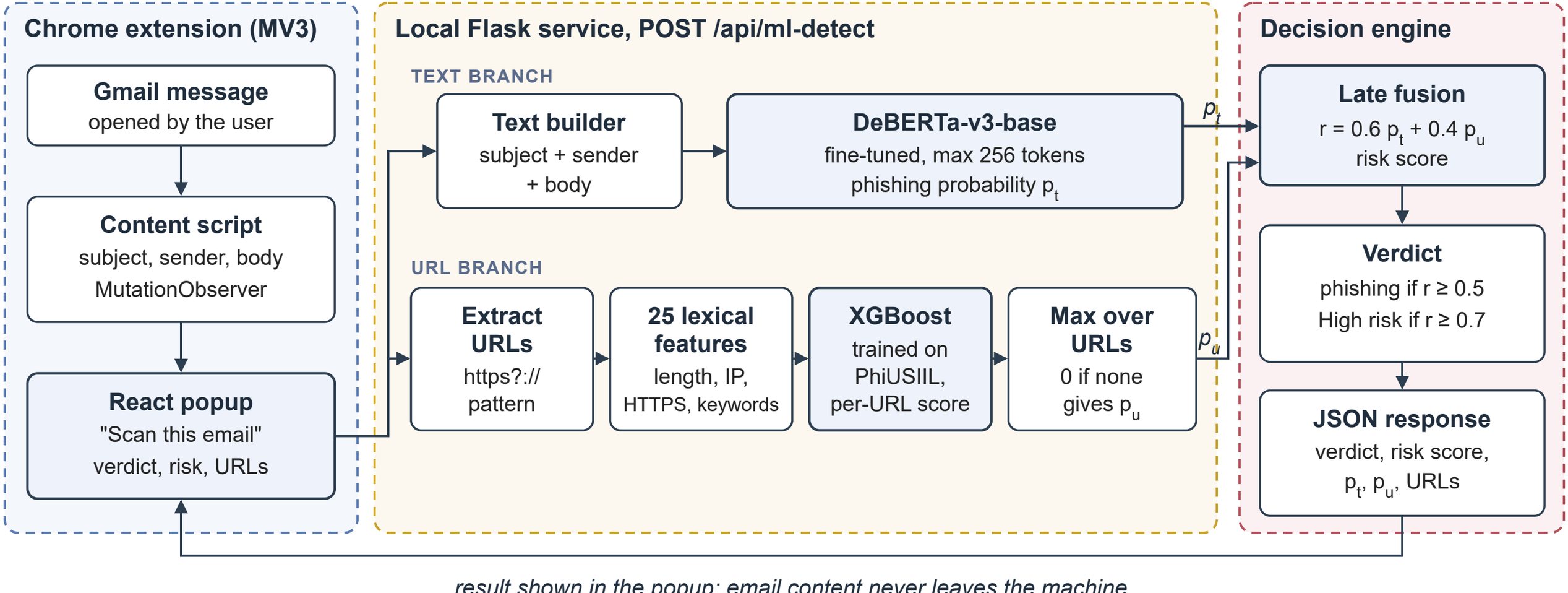

Chrome extension (MV3)
Gmail message
opened by the user
Content script
subject, sender, body
MutationObserver
React popup
"Scan this email"
verdict, risk, URLs
Local Flask service, POST /api/ml-detect
TEXT BRANCH
Text builder
subject + sender
+ body
DeBERTa-v3-base
fine-tuned, max 256 tokens
phishing probability p_t
p_t
URL BRANCH
Extract URLs
https?:// pattern
25 lexical features
length, IP, HTTPS, keywords
XGBoost
trained on PhiUSIIL, per-URL score
Max over URLs
0 if none
gives p_u
p_u
Decision engine
Late fusion
$r = 0.6\ p_t + 0.4\ p_u$
risk score
Verdict
phishing if $r \geq 0.5$
High risk if $r \geq 0.7$
JSON response
verdict, risk score,
p_t, p_u, URLs
result shown in the popup; email content never leaves the machine